

Ahnaf Shahriar

[Email](#) | [LinkedIn](#) | [Github](#)

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science in Computer Engineering

Sept. 2021 – Apr. 2026

- Recipient of Richard & Elizabeth Madter Entrance Scholarship and President's Scholarship of Distinction
- **Relevant Courses:** Computer Architecture(Verilog), Computer Networks, Systems Programming and Concurrency, Embedded Microprocessing Systems, Analog Control Systems, Compilers(Java), Numerical Methods

EXPERIENCE

Wireless Firmware Developer

Jan. 2025 – Apr. 2025

ON Semiconductor Corporation

Waterloo, ON

- **SDK Development:** Integrated *I3C* support and *BLE* profile services into sample code.
- **Wireless Connectivity:** Developed and integrated cutting-edge *LE Audio*, *HFP*, and *A2DP* functionality into an all-in-one Application.
- **Wireless Manager:** Modularized connectivity and implemented a state machine to optimize power efficiency by 25% and ease of use.
- **Test Automation:** Automated SDK test cases, reducing manual testing efforts by up to 80%

IC Design and Verification Intern

May 2023 – Aug. 2023, Jan. 2024 – May 2024

NXP Semiconductors Canada

Kanata, ON

- **IP Design:** Designed multiple IP blocks including *proprietary ECC IP* for dataplane processing chips.
- **Timing Analysis:** Spearheaded critical path improvements in IP block to increase Clock Frequency by 50%.
- **Functional Testing:** Designed brand new End-to-End functional tests in simulating *10GBps+* traffic for IP.
- **Unit Test Planning:** Created Simulation scenarios for testing High speed Dataplane Processing features.

Embedded Software Engineering Intern

Sept. 2022 – Dec. 2022

Synapse Product Development

Seattle, WA

- **Prototyping:** Leveraged Zephyr RTOS to create a proof of concept on *NRF52 BLE* device.
- **Python APIs:** Developed company specific lab automation software for equipment from *Agilent*, *Keysight*, *NI*, *Tektronik*.
- **Automation:** Streamlined testing and in house procedures using *Python* and *Bash*.
- **Driver Development:** Designed and implemented *drivers* of automated PCB testing Device(*I2C*, *UART*)

Firmware developer

Jan. 2022 – April 2022

Ford Motor Company of Canada

Remote

- **Unity/Cmock Test framework:** Lead developer for optimization for unit testing, achieving up to 30% *faster* runtime while using 50% less manually written test cases.
- **Automation:** Improved *Jenkins CI/CD* pipelines to support unit testing automation using *Python* for Linux server.
- **Embedded Trace Debugging:** Developed Parsing tools to debug through CAN and Serial message parsing of *10 million+* lines.
- **Automotive Design:** Maintained *AUTOSAR* standard design with *ISO26262 safety design* using *Davinci Configurator*.

PROJECTS

LC-3 Emulator: A C emulator for an educational ISA. Improves by 25% on research paper using *Python* data logging.

Real Time Executable: A RTOS implementation in STM32 capable of Pre-emptive task switching and its own Malloc

Stereo System: An embedded C implementation of a stereo playback system. Created with Quartus on Artix FPGA.

VHDL Compiler: A Java Compiler for creating combinational VHDL circuits. Using a boolean intermediate representation

TECHNICAL SKILLS

Languages: C/C++, Java, Python, Tcl, Bash scripting, ASM, VHDL, SystemVerilog/Verilog

Tools: Keil, Quartus, Git, Linux, Qemu, LLDB/GDB, Docker, WireShark, UVM, Matlab

Hardware: Oscilloscopes, Logic Analyzer, Multimeters, Spectrum Analyzer

Protocols: TCP/IP, JTAG, Serial, Ethernet, CAN/CAN-FD, LIN